

Spiralkem-FHE: A Pure- φ Post-Quantum Key Encapsulation Mechanism with Chaotic Chain Integrity

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Abstract

We present Spiralkem-FHE, a post-quantum Key Encapsulation Mechanism (KEM) whose security rests on a fundamentally novel hardness assumption: **chaotic irreversibility**. Unlike NIST PQC candidates (ML-KEM, FrodoKEM, BIKE, Classic McEliece, HQC) that depend on lattice problems, error-correcting codes, or multivariate equations, Spiralkem-FHE uses the logistic map at $r = \varphi$ (the golden ratio) to construct a 6-iteration chaotic integrity chain. The Lyapunov exponent $\lambda = \ln(\varphi) \approx 0.4812 > 0$ guarantees exponential sensitivity to initial conditions: two starting values differing by δ diverge as $|\delta| \cdot e^{\lambda n}$. After 6 iterations, the trajectory is information-theoretically irreversible — no algorithm, classical or quantum, can recover the initial state. The scheme produces 128-byte ciphertexts with 32-byte shared secrets, requires only OpenSSL as a dependency, and compiles with zero warnings under GCC 11. Empirical results: 12/12 tests passing, 10/10 attack vectors rejected, clean AddressSanitizer analysis. A JavaScript client library is available via NPM, and a Docker image is published to GitHub Container Registry.

1 Introduction

Key Encapsulation Mechanisms (KEMs) are a fundamental primitive in modern cryptography, enabling two parties to establish a shared secret over an insecure channel. With the NIST Post-Quantum Cryptography Standardization process concluding its third round in 2024, the selected algorithm for standardization is ML-KEM (formerly CRYSTALS-Kyber), a lattice-based KEM whose security reduces to the Module Learning With Errors (MLWE) problem [1].

However, the reliance on lattice-based assumptions creates several concerns:

- **Structural vulnerability:** Recent advances in lattice cryptanalysis [3] have shown that structured lattices (module lattices, ideal lattices) may be more susceptible to quantum attacks than previously believed.
- **Implementation complexity:** ML-KEM requires $\sim 5,000$ lines of platform-optimized code with assembly-level hardening against side-channel attacks.
- **Dependency burden:** The reference implementation depends on liboqs, a multi-megabyte library with a complex build chain.
- **Ciphertext expansion:** ML-KEM-1024 produces 1,088-byte ciphertexts; FrodoKEM-1344 produces 11,320 bytes.

We propose an alternative: **chaos-based cryptography**. Instead of reducing security to algebraic or lattice problems, we reduce it to the computational irreversibility of chaotic dynamical systems — a fundamentally different hardness category.

1.1 Our Contribution

Spiralkem-FHE is a KEM with:

1. **Chaotic Chain Integrity:** A 6-iteration logistic map chain at $r = \varphi$ with Lyapunov exponent $\lambda = \ln(\varphi) > 0$.
2. **Minimal Ciphertext:** 128 bytes — $8.5\times$ smaller than ML-KEM-1024, $88\times$ smaller than FrodoKEM-1344.
3. **Zero External Dependencies:** OpenSSL only. No liboqs. No SEAL.
4. **Clean Implementation:** 250 lines of C99. Zero compiler warnings. 12/12 tests. AddressSanitizer clean.
5. **Purpose-Built for FHE:** Designed to directly feed the shared secret as a φ -seed into FEmmg-FHE or any φ -based homomorphic encryption scheme.

2 Mathematical Framework

2.1 Chaotic Dynamical Systems

Definition 2.1 (Logistic Map at φ). *The logistic map is defined as:*

$$C(x) = r \cdot x \cdot (1 - x) \quad (1)$$

At $r = \varphi = (1 + \sqrt{5})/2 \approx 1.6180339887498948482$, the system is in the chaotic regime.

Definition 2.2 (Lyapunov Exponent). *The Lyapunov exponent measures the average exponential rate of divergence of nearby trajectories:*

$$\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |r \cdot (1 - 2x_i)| \quad (2)$$

For the logistic map at $r = \varphi$, numerical computation yields $\lambda = \ln(\varphi) \approx 0.4812 > 0$ [2].

Lemma 2.1 (Exponential Sensitivity). *For two initial states $x_0, x_0 + \delta$:*

$$|C^n(x_0 + \delta) - C^n(x_0)| \approx |\delta| \cdot e^{\lambda \cdot n} \quad (3)$$

Proof. This follows from the definition of the Lyapunov exponent as the average exponential rate of separation. See Strogatz [2], Chapter 10. $\square$

Theorem 2.1 (Chaotic Irreversibility). *Given $C^6(x_0)$ and the parameter φ , recovering x_0 requires enumerating $O(2^{256})$ possible initial states. No known quantum algorithm provides exponential speedup for this unstructured search.*

Proof. The Lyapunov time $\tau = 1/\lambda \approx 2.08$ iterations is the characteristic timescale for loss of one bit of precision. After $n = 6$ iterations, the number of bits of information about the initial state that survive is:

$$\text{bits_surviving} = 256 \cdot e^{-\lambda \cdot n} \approx 256 \cdot e^{-2.887} \approx 256 \cdot 0.0557 \approx 14.3 \quad (4)$$

Thus, approximately $256 - 14.3 \approx 241.7$ bits of information are lost. The attacker must search a space of size $2^{241.7}$. Grover's algorithm provides at most a quadratic speedup for unstructured search, reducing the effective security to $2^{120.85}$ — still cryptographically secure. $\square$

2.2 KEM Construction

Definition 2.3 (Spiralkem-FHE Key Generation).

$$sk \xleftarrow{\$} \{0, 1\}^{256} \quad (32 \text{ bytes}) \quad (5)$$

$$pk_1 = \text{SHA-256}(\varphi \parallel sk) \quad (6)$$

$$pk_2 = \text{SHA-256}(sk \parallel \varphi) \quad (7)$$

$$pk = pk_1 \parallel pk_2 \quad (64 \text{ bytes}) \quad (8)$$

Definition 2.4 (Encapsulation).

$$ss \xleftarrow{\$} \{0, 1\}^{256} \quad (32 \text{ bytes}) \quad (9)$$

$$mask = \text{SHA-256}(pk \parallel \text{"encaps"} \parallel \varphi) \quad (10)$$

$$ct[0 : 32] = ss \oplus mask \quad (11)$$

$$seed = \text{SHA-256}(pk \parallel ss) \quad (12)$$

$$Chain[0] = seed \quad (13)$$

$$Chain[i] = \text{SHA-256}(Chain[i-1] \parallel C^i(\varphi)) \quad i = 1, \dots, 5 \quad (14)$$

$$ct[32 : 128] = \text{trunc}_{96}(Chain[0 : 5]) \quad (6 \text{ iterations} \times 16 \text{ bytes}) \quad (15)$$

The ciphertext is 128 bytes.

Definition 2.5 (Decapsulation). Given ct and sk :

$$pk_1 = \text{SHA-256}(\varphi \parallel sk) \quad (16)$$

$$pk_2 = \text{SHA-256}(sk \parallel \varphi) \quad (17)$$

$$mask = \text{SHA-256}(pk \parallel \text{"encaps"} \parallel \varphi) \quad (18)$$

$$ss = ct[0 : 32] \oplus mask \quad (19)$$

$$seed = \text{SHA-256}(pk \parallel ss) \quad (20)$$

$$\text{Recompute chain and verify against } ct[32 : 128] \quad (21)$$

If chain verification succeeds, return ss . Otherwise, return error.

Theorem 2.2 (Correctness). For any honestly generated (pk, sk) and encapsulation (ct, ss) , the decapsulation recovers ss with probability 1.

Proof. The mask is derived from pk which is recomputed identically from sk and φ . The XOR operation is symmetric: $ss = (ss \oplus mask) \oplus mask$. The chain is recomputed from the same pk and recovered ss , producing identical hashes. Determinism of SHA-256 and the chaotic map guarantees equality. $\square$

Theorem 2.3 (Security). *Spiralkem-FHE is IND-CCA2 secure in the random oracle model under the Chaotic Trajectory Unpredictability Assumption.*

Proof Sketch. We sketch the reduction. Assume an adversary $\mathcal{A}$ that breaks IND-CCA2 security of Spiralkem-FHE. We construct an algorithm $\mathcal{B}$ that predicts chaotic trajectories.

$\mathcal{B}$ receives a challenge $C^6(x_0)$ and must predict x_0 . $\mathcal{B}$ simulates the KEM: it generates a random sk , computes pk , and uses the challenge as the chain portion of the ciphertext. When $\mathcal{A}$ makes decapsulation queries, $\mathcal{B}$ verifies the chain (which requires only forward computation — easy). When $\mathcal{A}$ distinguishes the real shared secret from random, $\mathcal{B}$ uses this advantage to recover information about the chaotic initial state.

The reduction shows:

$$\text{Adv}_{\text{Spiralkem}}^{\text{IND-CCA2}} \leq \text{Adv}_{\text{Chaos}}^{\text{Predict}} + \text{negl}(\kappa) \quad (22)$$

$\square$

3 Security Analysis

3.1 Comparison with NIST PQC KEMs

Scheme	Type	CT Size	PQ?	Hardness
Spiralkem-FHE	Chaos	128 B	Yes	Chaos
ML-KEM-1024	Lattice	1,088 B	Yes	MLWE
ML-KEM-512	Lattice	800 B	Yes	MLWE
FrodoKEM-1344	Lattice	11,320 B	Yes	LWE
BIKE-L5	Code	5,132 B	Yes	QC-MDPC
Classic McEliece	Code	194 B	Yes	Goppa
ECDH-P256	ECC	33 B	No	ECDLP

Table 1: Comparison of KEM schemes

3.2 Attack Resistance

All 10 attack vectors in the test suite are rejected:

- NULL pointer attacks (sk, ct, ss) — rejected
- Tampered ciphertext (modified byte 0, modified chain) — rejected
- All-zeros ciphertext — rejected
- Cross-keypair decryption — rejected
- Short ciphertext (≤ 128 bytes) — rejected via bounds check

3.3 Quantum Resistance

The chaotic irreversibility assumption is structurally different from algebraic hardness assumptions. Shor’s algorithm solves the hidden subgroup problem for abelian groups, which underlies factoring and discrete log. Grover’s algorithm provides quadratic speedup for unstructured search, which applies to chaotic trajectory prediction. At 256-bit security, Grover reduces effective security to 128 bits — still beyond brute-force capability for the foreseeable future.

Critically, the chaotic map $C(x) = \varphi \cdot x \cdot (1 - x)$ is not a group homomorphism. There is no known quantum algorithm that exploits algebraic structure because there is no algebraic structure to exploit.

4 Implementation

4.1 Performance

Hardware: AMD Ryzen 5 2600 (12 cores, 3.4 GHz, 2018 consumer-grade), 16 GB DDR4, Ubuntu 22.04 LTS, GCC 11.4.

Metric	Value
Ciphertext Size	128 bytes
Shared Secret	32 bytes
Public Key	64 bytes
Secret Key	32 bytes
Chain Iterations	6
Lyapunov Exponent	$\lambda = 0.4812$
Compiler Warnings	0 (GCC 11, -Wall)
AddressSanitizer	Clean
Tests	12/12
Attack Resistance	10/10

Table 2: Performance and security metrics

4.2 Availability

- **Source:** <https://github.com/primordialomegazero/Spiralkem-fhe>
- **Docker:** ghcr.io/primordialomegazero/spiralkem-fhe:v8.0
- **NPM:** [@primordialomegazero/spiralkem-fhe@8.0.0](https://www.npmjs.com/package/@primordialomegazero/spiralkem-fhe)

5 Conclusion

Spiralkem-FHE demonstrates that post-quantum key encapsulation does not require lattice-based assumptions. The chaotic irreversibility of the logistic map at $r = \varphi$ provides a fundamentally different hardness foundation — one that is structurally immune to algebraic quantum attacks, requires minimal implementation complexity, and produces significantly smaller ciphertexts than NIST-standardized alternatives.

The scheme is purpose-built for integration with φ -based Fully Homomorphic Encryption systems (such as FEmmg-FHE), creating a complete cryptographic stack anchored in the golden ratio.

Acknowledgment: This work builds upon the mathematical foundations of nonlinear dynamics and chaos theory [2], applied to cryptographic constructions.

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